

Python & C Programming

Task 01 — System Information & Basic Automation

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Part A: Environment Setup

Install the required tools and verify the Python installation.

Step 1: Install Python 3.x

1. Go to <https://www.python.org/downloads/>
2. Download the latest Python 3.x installer for your OS (Windows / macOS / Linux).
3. Run the installer. On Windows, make sure to check '**Add Python to PATH**' before clicking Install.

Step 2: Install VS Code (Recommended)

1. Go to <https://code.visualstudio.com/>
2. Download and install VS Code for your operating system.
3. Open VS Code -> Extensions (Ctrl+Shift+X) -> Search 'Python' -> Install the Microsoft Python extension.

Step 3: Verify Python Installation

Open your terminal (Command Prompt / PowerShell / Terminal) and run:

```
python --version
```

Expected Output:

```
Python 3.x.x
```

Take a screenshot of this output and save it in your submission folder.

Part B: system_info.py

Displays personal details and current date/time using the datetime module.

```
import datetime

name      = "Nawrin Zawna A N"
college   = "Kristu Jayanti College"
branch    = "CSE"
internship = "Cyber Security Summer Internship"
current_datetime = datetime.datetime.now().strftime("%Y-%m-%d %H:%M:%S")

print(f"Name: {name}")
print(f"College: {college}")
print(f"Branch: {branch}")
print(f"Internship: {internship}")
print(f>Date & Time: {current_datetime}")
```

Expected Output:

```
Name: Nawrin Zawna A N
College: Kristu Jayanti College
```

Branch: CSE
Internship: Cyber Security Summer Internship
Date & Time: 2026-06-08 14:32:00

Part C: user_input.py

Takes user input for name, age, and favourite technology, then prints a formatted summary.

```
name = input("Enter your name: ")
age = input("Enter your age: ")
fav_tech = input("Enter your favorite technology: ")

print(f"\nHello {name}!")
print(f"You are {age} years old.")
print(f"Your favorite technology is {fav_tech}.")
```

Example Output:

```
Enter your name: Nawrin Zawna A N
Enter your age: 20
Enter your favorite technology: Cyber Security

Hello Nawrin Zawna A N!
You are 20 years old.
Your favorite technology is Cyber Security.
```

Part D: calculator.py

A basic calculator that performs addition, subtraction, multiplication, and division on two user-entered numbers. Handles division by zero.

```
num1 = float(input("Enter First Number: "))
num2 = float(input("Enter Second Number: "))

print(f"Addition:      {num1 + num2}")
print(f"Subtraction:   {num1 - num2}")
print(f"Multiplication: {num1 * num2}")

if num2 != 0:
    print(f"Division:      {num1 / num2}")
else:
    print("Division: Cannot divide by zero!")
```

Example Output:

```
Enter First Number: 10
Enter Second Number: 5
Addition:      15.0
Subtraction:   5.0
Multiplication: 50.0
Division:      2.0
```

Part E: system_info_challenge.py

Displays OS name, current username, working directory, and Python version using the os, platform, and sys modules.

```
import os
```

```
import platform
import sys

print(f"Operating System:      {platform.system()} {platform.release()}")
print(f"Current Username:      {os.getlogin()}")
print(f"Current Working Directory: {os.getcwd()}")
print(f"Python Version:          {sys.version}")
```

Example Output:

```
Operating System:      Windows 11
Current Username:      Nawrin
Current Working Directory: C:\Users\Nawrin\Python_Task_01
Python Version:        3.11.4 (main, Jul 5 2023)
```

README.md

```
# Python_Task_01_NawrinZawnaAN
```

```
## Part B - system_info.py
```

Displays personal details and current date/time using the `datetime` module.

```
## Part C - user_input.py
```

Takes user input (name, age, favorite technology) and prints a formatted summary.

```
## Part D - calculator.py
```

Takes two numbers and performs add, subtract, multiply, divide.

Handles division by zero gracefully.

```
## Part E - system_info_challenge.py
```

Displays OS name, username, working directory, and Python version using `os`, `platform`, and `sys` modules.

```
## How to Run
```

```
python system_info.py
```

```
python user_input.py
```

```
python calculator.py
```

```
python system_info_challenge.py
```